

Advanced Financial Modeling Exercises

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Questions

1. Interest Rate Forwards

Suppose that we are in a single curve framework, the term structure of interest rates is defined by the following function:

$$L(0, t) = 1\% + 2\%(e^{-t/3})$$

Assume that at time $t = 0$ you enter a payer FRA with notional $100K$, the rate reset in $t = 2$ and flows are exchanged in $t = 3$.

- (a) Suppose that, as soon as we execute the trade all the term structure has a parallel shift of $+1\%$. Compute the value of the FRA after the shift
- (b) Did you make or lose money? Motivate the answer.

Solution: We are given:

$$L(0, t) = 0.01 + 0.02e^{-t/3}$$

Notional $N = 100,000$, accrual period $[2, 3]$, hence $\delta = 1$.

Step 1: Compute discount factors

Assuming simple compounding:

$$P(0, t) = \frac{1}{1 + tL(0, t)}$$

At $t = 2$:

$$L(0, 2) = 0.01 + 0.02e^{-2/3} \approx 0.020268$$

$$P(0, 2) = \frac{1}{1 + 2 \cdot 0.020268} \approx 0.961043$$

At $t = 3$:

$$L(0, 3) = 0.01 + 0.02e^{-1} \approx 0.017358$$

$$P(0, 3) = \frac{1}{1 + 3 \cdot 0.017358} \approx 0.950505$$

Step 2: Initial FRA rate

$$K = F(0; 2, 3) = \frac{P(0, 2)}{P(0, 3)} - 1$$

$$K \approx \frac{0.961043}{0.950505} - 1 \approx 0.011087 \quad (1.1087\%)$$

Step 3: Parallel shift of +1%

$$L^{new}(0, t) = L(0, t) + 0.01$$

$$L^{new}(0, 2) \approx 0.030268, \quad L^{new}(0, 3) \approx 0.027358$$

$$P^{new}(0, 2) = \frac{1}{1 + 2 \cdot 0.030268} \approx 0.942919$$

$$P^{new}(0, 3) = \frac{1}{1 + 3 \cdot 0.027358} \approx 0.924152$$

Step 4: New forward rate

$$F^{new}(0; 2, 3) = \frac{P^{new}(0, 2)}{P^{new}(0, 3)} - 1$$

$$F^{new}(0; 2, 3) \approx 0.020307 \quad (2.0307\%)$$

Step 5: Value of the payer FRA

$$V(0) = N \cdot \delta \cdot P^{new}(0, 3) \cdot (F^{new} - K)$$

$$V(0) = 100000 \cdot 1 \cdot 0.924152 \cdot (0.020307 - 0.011087)$$

$$V(0) \approx 852.08$$

Answer:

$V(0) \approx +852.08$

Interpretation:

The value is positive, so we made money. As a payer FRA, we pay the fixed rate K and receive the floating rate. After the upward shift, the forward rate increases, so we benefit from receiving a higher floating rate than the fixed rate we pay.

2. Interest Rate Swaps

Suppose that we are in a single curve framework, the term structure of interest rates is defined by the following function:

$$L(0, t) = 1.3\% + 1.1\%(t)$$

you enter a 3y receiver vanilla swap 1y forward, fixed and floating payments are annual, and the notional is 100 mln.

- (a) what is the fixed rate of the swap?
- (b) what is the relation between FRAs and Swaps? In this case, can you say how single forwards NPV are distributed across the life of the swap?
- (c) Is the position short or long interest rates? What is the difference with the multi curve framework?
- (d) Suppose that, as soon as we execute the trade, all the swap rate curve has a parallel shift of -1% . Compute the value of the derivative after the shift.

Solution:

We are given the term structure:

$$L(0, t) = 0.013 + 0.011t$$

and a 3-year receiver swap starting in 1 year, with annual payments and notional

$$N = 100,000,000.$$

The payment dates are:

$$T_0 = 1, \quad T_1 = 2, \quad T_2 = 3, \quad T_3 = 4.$$

Assuming a single-curve framework with simple compounding:

$$P(0, t) = \frac{1}{1 + tL(0, t)}.$$

Step 1: Discount factors

$$P(0, 1) = 0.9765625, \quad P(0, 2) = 0.934579, \quad P(0, 3) = 0.878735, \quad P(0, 4) = 0.814332.$$

(a) Fixed rate of the swap

The swap annuity is:

$$A(0) = P(0, 2) + P(0, 3) + P(0, 4) = 2.627646.$$

The forward swap rate is:

$$K = \frac{P(0, 1) - P(0, 4)}{A(0)} = \frac{0.9765625 - 0.814332}{2.627646} \approx 0.061740.$$

$K \approx 6.174\%$

(b) Relation between FRAs and swaps

A vanilla swap can be replicated as a portfolio of forward rate agreements (FRAs) on the periods:

$$[1, 2], \quad [2, 3], \quad [3, 4].$$

The corresponding forward rates are:

$$F(0; 1, 2) \approx 4.492\%, \quad F(0; 2, 3) \approx 6.355\%, \quad F(0; 3, 4) \approx 7.909\%.$$

Since the curve is upward sloping:

$$F(0; 1, 2) < K < F(0; 2, 3) < F(0; 3, 4).$$

Therefore, for a receiver swap:

- the first FRA has positive NPV ($K > F(0; 1, 2)$),
- the later FRAs have negative NPV ($K < F(0; 2, 3)$, $K < F(0; 3, 4)$).

The individual NPVs sum to zero at inception.

(c) Interest rate exposure and multi-curve framework

A receiver swap (receive fixed, pay floating) benefits from a decrease in interest rates, since the fixed leg becomes relatively more valuable. Therefore:

$$\text{Receiver swap} = \text{short interest rates (long duration)}$$

In a single-curve framework:

- the same curve is used for discounting and forward rates,
- the floating leg simplifies to

$$PV_{\text{float}} = N(P(0, T_0) - P(0, T_3)).$$

In a multi-curve framework:

- discounting and forwarding use different curves,
- the floating leg must be computed as the sum of discounted forward coupons,
- it no longer simplifies to a difference of discount factors.

(d) Value after a parallel shift of -1%

Since the exercise specifies a parallel shift of the swap rate curve, we can use the swap annuity as sensitivity:

$$V_{\text{receiver}} \approx N \cdot A(0) \cdot (K - S_{\text{new}})$$

A -1% shift implies:

$$S_{\text{new}} = K - 1\% \quad \Rightarrow \quad K - S_{\text{new}} = 1\% = 0.01.$$

Therefore:

$$V_{\text{receiver}} \approx 100,000,000 \times 2.627646 \times 0.01$$

$$V_{\text{receiver}} \approx 2,627,646.$$

$$V \approx +2.628 \text{ million}$$

The value is positive because the position is a receiver swap and interest rates have decreased.

3. Overnight Swaps

Starting from the definition of overnight rate in the EU:

$$r_{ois}(t, T) = \left[\prod_{i=t}^{T-\frac{1}{365}} \left(1 + \frac{r_{ON_i}}{360} \right) - 1 \right]$$

with the corresponding discount factors given by $P_d(t, T_i)$:

$$\begin{aligned} P_d(t, T_i) &= e^{-\int_t^{T_i} r_{ois}(s, s+1d) ds} \\ &= e^{-r_{ois}(t, T_i)(T_i-t)} \end{aligned}$$

- What is an Overnight Indexed Swap (OIS), and how is it defined?
- How can you price the floating leg and fixed leg of an OIS? Explain the methodology for pricing the floating leg based on realized overnight rates and the fixed leg based on predetermined fixed rates.
- Under what conditions can the floating leg equation be simplified? How?
- Define the annuity factor.

Solution:

(a) Definition of an Overnight Indexed Swap (OIS)

An Overnight Indexed Swap (OIS) is an interest rate swap in which:

- one leg pays a *fixed rate*,
- the other leg pays a *floating rate* linked to the daily overnight rate, compounded over each accrual period.

In the euro area, the overnight reference rate is based on the overnight market (historically EONIA, now STR-based conventions). Over each accrual period $[T_{i-1}, T_i]$, the floating payment is determined by compounding the realized overnight rates observed during that period.

Hence, the OIS exchanges:

- a predetermined fixed payment,
- against a compounded overnight payment.

(b) Pricing the floating leg and fixed leg of an OIS

Consider an OIS with payment dates $T_1, \dots, T_n$, accrual fractions $\delta_i = T_i - T_{i-1}$, notional N , and fixed rate K .

Floating leg

For each period $[T_{i-1}, T_i]$, the floating rate is given by the compounded overnight rate:

$$R_i^{\text{OIS}} = \frac{1}{\delta_i} \left(\prod_j (1 + r_{ON,j} \tau_j) - 1 \right),$$

where:

- $r_{ON,j}$ is the overnight rate observed on day j ,
- τ_j is the corresponding day-count fraction (typically 1/360 or 1/365, depending on convention).

Therefore, the floating payment at T_i is:

$$N \delta_i R_i^{\text{OIS}} = N \left(\prod_j (1 + r_{ON,j} \tau_j) - 1 \right).$$

The present value of the floating leg is then:

$$PV_{\text{float}} = N \sum_{i=1}^n P_d(t, T_i) \delta_i R_i^{\text{OIS}}.$$

Fixed leg

The fixed leg pays, at each payment date T_i :

$$N \delta_i K.$$

Hence its present value is:

$$PV_{\text{fixed}} = N K \sum_{i=1}^n \delta_i P_d(t, T_i).$$

At inception, the fair OIS fixed rate K is such that:

$$PV_{\text{fixed}} = PV_{\text{float}}.$$

(c) When can the floating leg be simplified? How?

The floating leg can be simplified when:

- the same overnight discount curve is used both for discounting and forwarding,
- there is no credit/liquidity spread separation, i.e. in a single-curve OIS framework,
- the floating coupon is exactly the compounded overnight rate generated by the discounting curve.

Under these conditions, for one accrual period $[T_{i-1}, T_i]$, we have:

$$1 + \delta_i R_i^{\text{OIS}} = \frac{P_d(t, T_{i-1})}{P_d(t, T_i)}.$$

Therefore,

$$\delta_i R_i^{\text{OIS}} = \frac{P_d(t, T_{i-1})}{P_d(t, T_i)} - 1.$$

Multiplying by $P_d(t, T_i)$, the discounted floating coupon becomes:

$$P_d(t, T_i) \delta_i R_i^{\text{OIS}} = P_d(t, T_{i-1}) - P_d(t, T_i).$$

Summing over all periods:

$$PV_{\text{float}} = N \sum_{i=1}^n (P_d(t, T_{i-1}) - P_d(t, T_i)).$$

This is a telescopic sum, so it simplifies to:

$$PV_{\text{float}} = N (P_d(t, T_0) - P_d(t, T_n)).$$

Thus, in the standard OIS framework, the floating leg can be written as:

$$PV_{\text{float}} = N (P_d(t, T_0) - P_d(t, T_n)).$$

(d) Definition of the annuity factor

The annuity factor is the present value of one unit paid on each fixed payment date, weighted by the corresponding accrual fractions. It is defined as:

$$A(t) = \sum_{i=1}^n \delta_i P_d(t, T_i).$$

It represents the present value of the fixed-leg payment stream per unit of fixed rate and per unit of notional.

Using the annuity factor, the fixed leg can be written compactly as:

$$PV_{\text{fixed}} = N K A(t).$$

Therefore, the fair fixed rate of the OIS is:

$$K = \frac{PV_{\text{float}}}{N A(t)}.$$

When the floating leg simplifies to

$$PV_{\text{float}} = N (P_d(t, T_0) - P_d(t, T_n)),$$

the fair OIS fixed rate becomes:

$$K = \frac{P_d(t, T_0) - P_d(t, T_n)}{A(t)}.$$

4. Convexity Adjustments

Assume a log-normal distribution for the underlying rate L :

$$dL_i(t) = \sigma L_i(t) dW(t)$$

- (a) Consider a basic interest rate payoff function, which pays a percentage of a notional N , and the percentage paid will be determined by the Libor rate $L(T_{i-1}, T_i)$. What happen if the payment date is different from the natural T_i ?
- (b) Solve the convexity equation:

$$cc(T_{i-1}, T_i) = P(t_0, T_i) E^{T_i} \left[\frac{L(T_{i-1}, T_i)}{P(T_{i-1}, T_i)} \right] - F(t_0; T_{i-1}, T_i)$$

hint: apply Ito's Lemma to $L_i^2(t)$, what can you say about the drift of this SDE?

Solution:

- a) If the payment date is different from the natural T_i , expectation of future libor rate won't be equal to the forward rate, we need to account for convexity.
- b) Recall the definition of the simple Libor rate $L(T_{i-1}, T_i)$:

$$P(T_{i-1}, T_i) = \frac{1}{1 + \tau_i L(T_{i-1}, T_i)}$$

Then the expectation inside the $cc(T_{i-1}, T_i)$ equation can be written as follows:

$$\begin{aligned} \mathbb{E}^{T_i} \left[\frac{L(T_{i-1}, T_i)}{P(T_{i-1}, T_i)} \right] &= \mathbb{E}^{T_i} [L(T_{i-1}, T_i) + \tau_i L^2(T_{i-1}, T_i)] \\ &= F(t_0, T_{i-1}, T_i) + \tau_i \mathbb{E}^{T_i} [L^2(T_{i-1}, T_i)] \end{aligned}$$

We have different choices to determine the expectation at the right-hand side in the above expression. A basic choice for the Libor rate is then to define it as a lognormal process, as follows

$$dL_i(t) = \sigma L_i(t) dW_i^i(t)$$

We want to solve for $L_i^2(t)$. Consider the dynamics

$$dL_i(t) = \sigma L_i(t) dW_i^i(t)$$

We want to solve for $L_i^2(t)$.

To solve this stochastic differential equation (SDE), we can use Ito's Lemma. First, let's apply Ito's Lemma to the function $f(L_i) = L_i^2$.

Ito's Lemma states that for a function $f(t, x)$ of a stochastic process $x(t)$ satisfying the stochastic differential equation

$$dx(t) = \mu(t, x)dt + \sigma(t, x)dW(t)$$

the corresponding stochastic process $f(t, x(t))$ satisfies the stochastic differential equation:

$$df(t, x(t)) = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW(t)$$

Applying this to our function $f(L_i) = L_i^2$, we have:

$$\begin{aligned} dL^2 &= \frac{\partial L^2}{\partial L} dL + \frac{1}{2} \frac{\partial^2 L^2}{\partial L^2} dL^2 \\ dL^2 &= 2L dL + \frac{1}{2} 2 dL^2 \\ dL^2 &= 2\sigma L^2 dW_i(t) + \sigma^2 L_i^2 dt \end{aligned}$$

The equation has a drift term, so it's not a martingale.
with the solution given by, using $t_0 = 0$,

$$L_i(T_{i-1}) = L_i(t_0) e^{-\frac{1}{2}\sigma^2 T_{i-1} + \sigma W_i^i(T_{i-1})}$$

Taking squares in the explicit solution of the lognormal Libor process, we obtain

$$L_i^2(T_{i-1}) = L_i^2(t_0) \exp \left(-\sigma^2 T_{i-1} + 2\sigma W_i^i(T_{i-1}) \right),$$

where, for simplicity, we set $t_0 = 0$.

Taking expectations under the T_i -forward measure gives

$$\mathbb{E}^{T_i}[L_i^2(T_{i-1})] = L_i^2(t_0) e^{-\sigma^2 T_{i-1}} \mathbb{E}^{T_i} \left[e^{2\sigma W_i^i(T_{i-1})} \right].$$

Since $W_i^i(T_{i-1}) \sim N(0, T_{i-1})$, we use the moment-generating function of a normal random variable:

$$\mathbb{E}[e^{aW(T)}] = e^{\frac{1}{2}a^2T}.$$

Setting $a = 2\sigma$, we get

$$\mathbb{E}^{T_i} \left[e^{2\sigma W_i^i(T_{i-1})} \right] = e^{\frac{1}{2}(2\sigma)^2 T_{i-1}} = e^{2\sigma^2 T_{i-1}}.$$

Therefore,

$$\mathbb{E}^{T_i}[L_i^2(T_{i-1})] = L_i^2(t_0) e^{-\sigma^2 T_{i-1}} e^{2\sigma^2 T_{i-1}} = L_i^2(t_0) e^{\sigma^2 T_{i-1}}.$$

Substituting this result into the convexity correction formula yields

$$cc(T_{i-1}, T_i) = P(t_0, T_i) \left(L_i(t_0) + \tau_i L_i^2(t_0) e^{\sigma^2 T_{i-1}} \right) - L_i(t_0).$$

In practice, this specification is not optimal, as it is not clear how to specify the parameter σ in the dynamics. One may consider to choose σ to be the caplet ATM volatility, however this is not a unique choice.

5. HJM Process

Define the HJM dynamics of $f(t, T)$ under the risk neutral measure $\mathbb{Q}$

$$df(t, T) = \left(\sigma_f(t, T) \int_t^T \sigma_f(t, u) du \right) dt + \sigma_f(t, T) dW_t$$

Recall the change of numeraire equality

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} = \frac{P(t, T)}{P(0, T)} \frac{B(0)}{B(t)}$$

- (a) Show that $f(t, T)$ is a martingale under the $\mathbb{Q}^T$ forward measure.
(b) What is the difference between forward measure $\mathbb{Q}^T$ and terminal forward measure $\mathbb{Q}^{T_f}$?

Solution:

- a) We want to show that $f(t, T)$ is a martingale under the T -forward measure $\mathbb{Q}^T$.
We start from the change of numeraire relation

$$\left. \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \right|_{\mathcal{F}_t} = \frac{P(t, T)}{P(0, T)} \frac{B(0)}{B(t)}.$$

Under the risk-neutral measure $\mathbb{Q}$, the bond price dynamics can be written as

$$\frac{dP(t, T)}{P(t, T)} = r_t dt + \sigma_P(t, T) dW_t.$$

Applying Itô's Lemma to $\ln P(t, T)$ gives

$$d \ln P(t, T) = \left(r_t - \frac{1}{2} \sigma_P^2(t, T) \right) dt + \sigma_P(t, T) dW_t.$$

Integrating from 0 to t , we obtain

$$\frac{P(t, T)}{P(0, T)} = \exp \left(\int_0^t \left(r_u - \frac{1}{2} \sigma_P^2(u, T) \right) du + \int_0^t \sigma_P(u, T) dW_u \right).$$

Since

$$\frac{B(0)}{B(t)} = \exp \left(- \int_0^t r_u du \right),$$

substituting into the Radon–Nikodym derivative yields

$$\left. \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \right|_{\mathcal{F}_t} = \exp \left(- \frac{1}{2} \int_0^t \sigma_P^2(u, T) du + \int_0^t \sigma_P(u, T) dW_u \right). \quad (1)$$

By Girsanov's theorem, if

$$\left. \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \right|_{\mathcal{F}_t} = \exp \left(- \frac{1}{2} \int_0^t y_u^2 du + \int_0^t y_u dW_u \right),$$

then the process

$$W_t^T = W_t - \int_0^t y_u du$$

is a Brownian motion under $\mathbb{Q}^T$, that is,

$$dW_t^T = dW_t - y_t dt.$$

Comparing this with (1), we identify

$$y_t = \sigma_P(t, T).$$

Therefore,

$$dW_t^T = dW_t - \sigma_P(t, T) dt.$$

In the HJM framework, the bond volatility is related to the forward rate volatility by

$$\sigma_P(t, T) = - \int_t^T \sigma_f(t, u) du.$$

Hence,

$$dW_t^T = dW_t + \int_t^T \sigma_f(t, u) du dt,$$

or equivalently,

$$dW_t = dW_t^T - \int_t^T \sigma_f(t, u) du dt.$$

Now substitute this into the HJM dynamics under $\mathbb{Q}$:

$$df(t, T) = \left(\sigma_f(t, T) \int_t^T \sigma_f(t, u) du \right) dt + \sigma_f(t, T) dW_t.$$

We obtain

$$\begin{aligned} df(t, T) &= \left(\sigma_f(t, T) \int_t^T \sigma_f(t, u) du \right) dt + \sigma_f(t, T) \left(dW_t^T - \int_t^T \sigma_f(t, u) du dt \right) \\ &= \sigma_f(t, T) dW_t^T. \end{aligned}$$

Therefore, the drift vanishes under the T -forward measure, so $f(t, T)$ is a martingale under $\mathbb{Q}^T$.

b) The difference between the forward measure $\mathbb{Q}^T$ and the terminal forward measure $\mathbb{Q}^{T_f}$ lies in the choice of numeraire.

- Under the T -forward measure $\mathbb{Q}^T$, the numeraire is the zero-coupon bond $P(t, T)$.
- Under the terminal forward measure $\mathbb{Q}^{T_f}$, the numeraire is the longest bond $P(t, T_f)$, where $T_f > T$.

By Girsanov's theorem, under the terminal forward measure we have

$$dW_t^{T_f} = dW_t - \sigma_P(t, T_f) dt.$$

Using again the HJM relation

$$\sigma_P(t, T_f) = - \int_t^{T_f} \sigma_f(t, u) du,$$

it follows that

$$dW_t^{T_f} = dW_t + \int_t^{T_f} \sigma_f(t, u) du dt,$$

or equivalently

$$dW_t = dW_t^{T_f} - \int_t^{T_f} \sigma_f(t, u) du dt.$$

Substituting this into the risk-neutral HJM dynamics gives

$$\begin{aligned} df(t, T) &= \left(\sigma_f(t, T) \int_t^T \sigma_f(t, u) du \right) dt + \sigma_f(t, T) \left(dW_t^{T_f} - \int_t^{T_f} \sigma_f(t, u) du dt \right) \\ &= \sigma_f(t, T) \left(\int_t^T \sigma_f(t, u) du - \int_t^{T_f} \sigma_f(t, u) du \right) dt + \sigma_f(t, T) dW_t^{T_f} \\ &= - \left(\sigma_f(t, T) \int_T^{T_f} \sigma_f(t, u) du \right) dt + \sigma_f(t, T) dW_t^{T_f}. \end{aligned}$$

Therefore:

- under $\mathbb{Q}$, the drift is

$$\sigma_f(t, T) \int_t^T \sigma_f(t, u) du;$$

- under $\mathbb{Q}^T$, the drift is zero, so $f(t, T)$ is a martingale;
- under $\mathbb{Q}^{T_f}$, the drift is

$$-\sigma_f(t, T) \int_T^{T_f} \sigma_f(t, u) du.$$

Hence, the T -forward measure is specifically associated with the maturity T of the forward rate under consideration, while the terminal forward measure is associated with a fixed final maturity T_f and is often used when modelling all forward rates up to a terminal horizon under a single measure.

6. Solve the Hull–White Process

Consider the one-factor Hull–White model (extended Vasicek) under the risk neutral measure $\mathbb{Q}$:

$$dr(t) = \kappa(\theta(t) - r(t))dt + \sigma dW_t, \quad r(0) = r_0,$$

where $\kappa > 0$ and $\sigma > 0$ are constants, while $\theta(t)$ is a deterministic function chosen to fit the initial term structure

$$\theta(t) = f^M(0, t) + \frac{1}{\kappa} \frac{\partial}{\partial t} f^M(0, t) + \frac{\sigma^2}{2\kappa^2} (1 - e^{-2\kappa t}),$$

where $f^M(0, t)$ denotes the market instantaneous forward rate at time 0 for maturity t .

- Solve the stochastic differential equation for $r(t)$ and derive the conditional expectation $\mathbb{E}^{\mathbb{Q}}[r(t) \mid \mathcal{F}_s]$ and conditional variance $\text{Var}^{\mathbb{Q}}[r(t) \mid \mathcal{F}_s]$, for $0 \leq s < t$.
- What is the difference between the Hull–White model and the Vasicek model?

Solution:

a) Define

$$X(t) = f(t, r(t)) := e^{\kappa t} r(t).$$

We apply Itô's lemma to the function $f(t, r) = e^{\kappa t} r$. Its derivatives are

$$\frac{\partial f}{\partial t}(t, r) = \kappa e^{\kappa t} r, \quad \frac{\partial f}{\partial r}(t, r) = e^{\kappa t}, \quad \frac{\partial^2 f}{\partial r^2}(t, r) = 0.$$

Since

$$dr(t) = \kappa(\theta(t) - r(t))dt + \sigma dW_t,$$

Itô's lemma gives

$$dX(t) = \frac{\partial f}{\partial t}(t, r(t))dt + \frac{\partial f}{\partial r}(t, r(t))dr(t) + \frac{1}{2} \frac{\partial^2 f}{\partial r^2}(t, r(t))\sigma^2 dt.$$

Because $\frac{\partial^2 f}{\partial r^2} = 0$, this becomes

$$dX(t) = \kappa e^{\kappa t} r(t) dt + e^{\kappa t} dr(t).$$

Substituting the dynamics of $r(t)$,

$$\begin{aligned} dX(t) &= \kappa e^{\kappa t} r(t) dt + e^{\kappa t} [\kappa(\theta(t) - r(t))dt + \sigma dW_t] \\ &= \kappa e^{\kappa t} \theta(t) dt + \sigma e^{\kappa t} dW_t. \end{aligned}$$

Hence,

$$d(e^{\kappa t} r(t)) = \kappa e^{\kappa t} \theta(t) dt + \sigma e^{\kappa t} dW_t.$$

Integrating from s to t ,

$$e^{\kappa t} r(t) - e^{\kappa s} r(s) = \int_s^t \kappa e^{\kappa u} \theta(u) du + \sigma \int_s^t e^{\kappa u} dW_u.$$

Therefore,

$$r(t) = e^{-\kappa(t-s)} r(s) + \kappa \int_s^t e^{-\kappa(t-u)} \theta(u) du + \sigma \int_s^t e^{-\kappa(t-u)} dW_u.$$

Taking conditional expectation with respect to $\mathcal{F}_s$, and using that the stochastic integral has zero conditional mean,

$$\mathbb{E}^{\mathbb{Q}}[r(t) \mid \mathcal{F}_s] = e^{-\kappa(t-s)} r(s) + \kappa \int_s^t e^{-\kappa(t-u)} \theta(u) du.$$

For the conditional variance, only the stochastic integral contributes. By Itô isometry,

$$\begin{aligned} \text{Var}^{\mathbb{Q}}[r(t) \mid \mathcal{F}_s] &= \sigma^2 \int_s^t e^{-2\kappa(t-u)} du \\ &= \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa(t-s)}). \end{aligned}$$

Thus, conditionally on $\mathcal{F}_s$, the short rate is Gaussian with mean

$$e^{-\kappa(t-s)} r(s) + \kappa \int_s^t e^{-\kappa(t-u)} \theta(u) du$$

and variance

$$\frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa(t-s)}).$$

b) The Hull–White model is an extended version of the Vasicek model.

In the Vasicek model, the short rate follows

$$dr(t) = \kappa(\theta - r(t))dt + \sigma dW_t,$$

where θ is constant. This implies that the model is not flexible enough to fit exactly a generic observed initial term structure.

In the Hull–White model, instead, the dynamics are

$$dr(t) = \kappa(\theta(t) - r(t))dt + \sigma dW_t,$$

where $\theta(t)$ is deterministic and time-dependent. This time dependence allows the model to match the initial term structure exactly.

Therefore:

- Vasicek has constant long-run mean θ ;
- Hull–White replaces it with a time-dependent function $\theta(t)$;
- Hull–White fits the initial yield curve exactly, while Vasicek generally does not;
- both models are Gaussian and analytically tractable.

7. Solve the G2++ Process

The Gaussian two factor model defines the short rate as the sum of two correlated Gaussian random variables, under the risk neutral measure $\mathbb{Q}$ the interest rate dynamics are

$$r(t) = x(t) + y(t) + \varphi(t), \quad r(0) = r_0.$$

With $\varphi(t)$ being a deterministic function of time, chosen to exactly fit the initial term structure of interest rates, $\varphi(0) = r_0$

$$\begin{aligned} \varphi(t) = & f(0, t) + \frac{\sigma^2}{2a^2}(1 - e^{-at})^2 \\ & + \frac{\eta^2}{2b^2}(1 - e^{-bt})^2 + \rho \frac{\sigma\eta}{ab}(1 - e^{-at})(1 - e^{-bt}), \end{aligned}$$

and the two processes $x(t)$ and $y(t)$ defined as

$$\begin{aligned} dx(t) &= -ax(t)dt + \sigma dW_1(t), & x(0) &= 0; \\ dy(t) &= -by(t)dt + \eta dW_2(t), & y(0) &= 0; \end{aligned}$$

where W_1 and W_2 are two different Brownian motion with correlation ρ

- (a) Compute conditional expectations $\mathbb{E}^{\mathbb{Q}}[x(t)|x(s)]$ and $\mathbb{E}^{\mathbb{Q}}[y(t)|y(s)]$, what is the value of r_0 ?
- (b) Compute factor variances and covariance.
- (c) What is the difference with the one factor model?

Solution: If we integrate we obtain the two stochastic differential equations solutions

$$\begin{aligned}x(t) &= x(s)e^{-a(t-s)} + \sigma \int_s^t e^{-a(t-u)} dW_1(u), \\y(t) &= y(s)e^{-b(t-s)} + \eta \int_s^t e^{-b(t-u)} dW_2(u);\end{aligned}$$

then the short rate evaluated at t_0 for a given maturity T is expressed as

$$r_0(T) = \sigma \int_0^T e^{-a(T-u)} dW_1(u) + \eta \int_0^T e^{-b(T-u)} dW_2(u) + \varphi(T);$$

letting $T \rightarrow \infty$ the two factors reverts to their initial value $x(\infty), y(\infty) \rightarrow 0$ and the $r(T)$ process mean reverts to $\varphi(t)$.

By the properties of stochastic integral of deterministic functions we obtain mean, variance and covariance from the bivariate Normal distribution of $r(t)$ conditional on $r(s)$ with $s < t$ as

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[x(t)|x(s)] &= x(s)e^{-a(t-s)}, \quad \mathbb{E}^{\mathbb{Q}}[y(t)|y(s)] = y(s)e^{-b(t-s)}; \\Var^{\mathbb{Q}}[x(t)] &= \frac{\sigma^2}{2a}[1 - e^{-2a(t-s)}], \quad Var_s^{\mathbb{Q}}[y(t)] = \frac{\eta^2}{2b}[1 - e^{-2b(t-s)}]; \\cov_s[x(t), y(t)] &= \rho \frac{\sigma\eta}{a+b}[1 - e^{-(a+b)(t-s)}].\end{aligned}$$

The two processes can also be rewritten in terms of two independent Brownian motions $\tilde{W}_1$ and $\tilde{W}_2$ applying a Cholesky decomposition

$$\begin{aligned}dW_1(t) &= d\tilde{W}_1, \\dW_2(t) &= \rho d\tilde{W}_1 + \sqrt{1 - \rho^2} d\tilde{W}_2.\end{aligned}$$

Substituting in the two equation for $x(t)$ and $y(t)$ by integration we get

$$\begin{aligned}x(t) &= x(s)e^{-a(t-s)} + \sigma \int_0^t e^{-a(t-u)} d\tilde{W}_1(u), \\y(t) &= y(s)e^{-b(t-s)} + \eta\rho \int_0^t e^{-b(t-u)} d\tilde{W}_1(u) + \eta\sqrt{1 - \rho^2} \int_0^t e^{-b(t-u)} d\tilde{W}_2(u).\end{aligned}$$

The main difference with the one-factor Gaussian model is that, in the G2++ model, the short rate is driven by **two** mean-reverting Gaussian factors instead of only one.

In the one-factor model, the short rate is typically written as

$$r(t) = x(t) + \varphi(t),$$

where $x(t)$ is a single Ornstein–Uhlenbeck process. In the G2++ model, instead,

$$r(t) = x(t) + y(t) + \varphi(t),$$

with two correlated Ornstein–Uhlenbeck factors.

The presence of two factors makes the model more flexible, since it can capture a richer set of movements of the yield curve. In particular:

- the one-factor model mainly generates parallel shifts of the term structure;
- the two-factor model can better reproduce changes in both **slope** and **curvature**;
- the correlation parameter ρ allows for a more realistic dependence structure across maturities.

Therefore, G2++ provides a better fit to the joint dynamics of interest rates than the one-factor model, while still preserving analytical tractability because both factors remain Gaussian.

8. xVA Modeling

- (a) Define briefly Counterparty Credit Risk and CVA. Show that, at valuation time t_0 , with $\tau > t_0$ (no default happened yet), the price of a derivative under counterparty risk is:

$$V_D(t_0) = V(t_0) - \mathbb{E}^Q [1_{\{\tau \leq T\}} \cdot LGD(\tau) \cdot D(t_0, \tau) \cdot V(\tau)^+ | \mathcal{F}_{t_0}].$$

- (b) Write the uCVA equation for an interest rate swap.
(c) Define netting and collateral. Which components of the collateral value are stochastic?
(d) Why does FVA exist? Explain the analogy between credit and funding spread in FVA and DVA.

Solution:

- a) **Counterparty Credit Risk** is the risk that the counterparty defaults before the final maturity of the contract and fails to fulfill its contractual obligations.

CVA (*Credit Valuation Adjustment*) is the expected loss due to counterparty default, and represents the difference between the risk-free value and the risky value of a derivative.

If the default of the counterparty occurs after the final maturity T , the derivative is unaffected by counterparty risk and its value is simply

$$1_{\{\tau > T\}} V(t_0).$$

If instead the default occurs before maturity, i.e. $\tau \leq T$, we distinguish three components:

1. all contractual cash flows up to the default time are exchanged, giving

$$1_{\{\tau \leq T\}} V(t_0, \tau),$$

where $V(t_0, \tau)$ denotes the value at time t_0 of the cash flows up to τ ;

2. if the exposure at default is positive, only a fraction $R(\tau)$ is recovered:

$$1_{\{\tau \leq T\}} R(\tau) \max(V(\tau), 0);$$

3. if the exposure is negative, the full amount must be paid:

$$1_{\{\tau \leq T\}} \min(V(\tau), 0).$$

Thus, the value of the *risky* derivative at time t_0 is

$$\begin{aligned} V_D(t_0) = \mathbb{E}^Q [& 1_{\{\tau > T\}} V(t_0) + 1_{\{\tau \leq T\}} V(t_0, \tau) \\ & + D(t_0, \tau) 1_{\{\tau \leq T\}} R(\tau) \max(V(\tau), 0) \\ & + D(t_0, \tau) 1_{\{\tau \leq T\}} \min(V(\tau), 0) \mid \mathcal{F}_{t_0}]. \end{aligned}$$

Using the identity $x = \max(x, 0) + \min(x, 0)$, we obtain

$$\begin{aligned} V_D(t_0) = \mathbb{E}^Q [& 1_{\{\tau > T\}} V(t_0) + 1_{\{\tau \leq T\}} V(t_0, \tau) \\ & + D(t_0, \tau) 1_{\{\tau \leq T\}} V(\tau) \\ & + D(t_0, \tau) 1_{\{\tau \leq T\}} (R(\tau) - 1) \max(V(\tau), 0) \mid \mathcal{F}_{t_0}]. \end{aligned}$$

The first three terms correspond to the risk-free valuation. Indeed,

$$\mathbb{E}^Q [1_{\{\tau > T\}} V(t_0) + 1_{\{\tau \leq T\}} V(t_0, \tau) + D(t_0, \tau) 1_{\{\tau \leq T\}} V(\tau)] = V(t_0).$$

Therefore, the value of the risky derivative becomes

$$\begin{aligned} V_D(t_0) &= V(t_0) + \mathbb{E}^Q [1_{\{\tau \leq T\}} (R(\tau) - 1) D(t_0, \tau) V(\tau)^+ \mid \mathcal{F}_{t_0}] \\ &= V(t_0) - \mathbb{E}^Q [1_{\{\tau \leq T\}} LGD(\tau) D(t_0, \tau) V(\tau)^+ \mid \mathcal{F}_{t_0}] \\ &= V(t_0) - \text{uCVA}(t_0). \end{aligned}$$

b) Starting from the uCVA equation:

$$\text{uCVA}(t) = 1_{\{\tau > t\}} \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{E}^Q \left[\left(e^{-\int_t^{t_{i-1}} \lambda(s) ds} - e^{-\int_t^{t_i} \lambda(s) ds} \right) LGD(t_{i-1}) \cdot D(t, t_{i-1}) \cdot PV^+(t_{i-1}) \mid \mathcal{F}_t \right]$$

Assume:

- a finite number of time steps N ;
- constant loss given default;
- independence between default intensities and interest rates;
- deterministic hazard rates at time t_0 .

Under these assumptions, the default probability over the interval $(t_{i-1}, t_i]$ is

$$\mathbb{Q}(t_{i-1} < \tau \leq t_i) = e^{-\int_{t_0}^{t_{i-1}} \lambda(s) ds} - e^{-\int_{t_0}^{t_i} \lambda(s) ds},$$

and $LGD(t_{i-1}) = LGD$ can be taken outside the expectation. Hence,

$$\text{uCVA}_{\text{swap}}(t_0) = LGD \sum_{i=1}^N \left(e^{-\int_{t_0}^{t_{i-1}} \lambda(s) ds} - e^{-\int_{t_0}^{t_i} \lambda(s) ds} \right) \mathbb{E}^Q [D(t_0, t_{i-1}) PV^+(t_{i-1}) \mid \mathcal{F}_{t_0}].$$

Therefore, the unilateral CVA for an interest rate swap can be written as

$$\text{uCVA}_{\text{swap}}(t_0) = LGD \sum_{i=1}^N \left(e^{-\int_{t_0}^{t_{i-1}} \lambda(s) ds} - e^{-\int_{t_0}^{t_i} \lambda(s) ds} \right) \mathbb{E}^Q [D(t_0, t_{i-1}) PV^+(t_{i-1}) \mid \mathcal{F}_{t_0}].$$

The quantity

$$\mathbb{E}^Q [D(t_0, t_{i-1}) PV^+(t_{i-1}) \mid \mathcal{F}_{t_0}]$$

is the discounted expected positive exposure of the swap. In practice, for an interest rate swap this term is often approximated through a portfolio of swaptions.

- c) **Netting** is a contractual agreement under which multiple transactions with the same counterparty are aggregated, so that in case of default only the net positive exposure is relevant rather than the sum of exposures trade by trade.

Collateral is a risk mitigation mechanism whereby one party posts assets, typically cash or securities, in order to reduce counterparty exposure.

The collateral value contains stochastic components. In particular:

- the mark-to-market value of the underlying portfolio is stochastic;
- the exposure process driving collateral calls is stochastic;
- the collateral account itself is stochastic, since it depends on the evolution of exposure;
- if non-cash collateral is posted, the market value of the collateral assets is stochastic;
- timing effects related to margin calls and the margin period of risk also affect the effective collateral value.

- d) **FVA** (*Funding Valuation Adjustment*) exists because a financial institution cannot in general fund itself at the risk-free rate. If a derivative position requires funding, especially when it is uncollateralized or imperfectly collateralized, the institution faces a funding cost that must be incorporated into valuation.

The analogy between credit and funding spread is the following:

- in **CVA** and **DVA**, the relevant spread is the *credit spread*, reflecting the default risk of the counterparty or of the institution itself;
- in **FVA**, the relevant spread is the *funding spread*, namely the spread between the institution's funding rate and the risk-free rate.

Hence:

- **CVA** represents the expected loss due to counterparty default;
- **DVA** represents the effect of the institution's own default risk;
- **FVA** represents the cost or benefit of funding the position through time.